ROS 2 Labs (Windows)

Seventeen hands-on ROS 2 (Jazzy) projects (sixteen are in the repository now; the Docker lab U5 follows shortly). **Every lab is a complete, separate project** in its own folder under `labs\`: its own environment (`pixi.toml`), source code, scripts, automated test and manual (`README.md`). You can do any lab on its own, in any order; nothing is shared between them except the tools you install once.

1. Install two tools (once per computer)

Open **PowerShell** and run, then close and reopen PowerShell:

```
winget install --id Git.Git -e --source winget
winget install --id prefix-dev.pixi -e --source winget
```

2. Get the labs

```
git clone <repository-url> ros-labs
cd ros-labs
```

Only want one lab? Download just that folder:

```
git clone --filter=blob:none --sparse <repository-url> ros-labs
cd ros-labs
git sparse-checkout set labs/Lab-02-robot-model
```

3. Run a lab

```
cd labs\Lab-02-robot-model
```

Open that folder's **`README.md`** (or its PDF in `pdf\`) and follow it. Every manual starts with the same two commands, `pixi install` (downloads the lab's environment, minutes the first time) and `pixi run verify`, and then continues with the lab's own steps. The last step of each manual is a one-command automated check (`pixi run test-...`).

The labs

Folder (under `labs\`)	Project	Also needs	Time
Course: build the robot step by step			
`Lab-01-turtle-catcher`	Turtle Catcher: nodes, topics, services, parameters, launch	C++ compiler	45 min
`Lab-02-robot-model`	DiffBot model: URDF, xacro, TF, RViz2		45 min
`Lab-03-simulation-odometry`	Simulator and wheel odometry		90 min
`Lab-04-lidar-avoidance`	LiDAR obstacle avoidance		60 min

Folder (under `Labs\`)	Project	Also needs	Time
`Lab-05-autonomous-navigation`	A* planner + pure pursuit + actions		90 min
`Lab-06-nav2`	The real Nav2 stack		45 min
Original public repositories, built and run unmodified (downloaded by the lab at a pinned commit)			
`Lab-U1-packt-ros2-from-scratch`	Packt "ROS 2 from Scratch", chapter 9 (Python + C++)	C++ compiler	90 min
`Lab-U2-turtlesim`	turtlesim built from source	C++ compiler	45 min
`Lab-U3-urdf-tutorial`	the eight `urdf_tutorial` robots		45 min
`Lab-U4-nav2-custom-planner`	Your own planner inside Nav2	C++ compiler	60 min
`Lab-U5-gazebo-slam-docker`	Gazebo, SLAM and Nav2 in Docker *(coming soon, not in the repository yet)*	Docker Desktop	30 min
NVIDIA Isaac Sim (physics simulator)			
`Lab-I1-isaac-drive`	Drive in Isaac Sim from ROS 2	NVIDIA RTX GPU	45 min
`Lab-I2-isaac-lidar`	LiDAR in Isaac Sim + obstacle avoider	NVIDIA RTX GPU	30 min
`Lab-I3-isaac-camera`	Camera: follow a red cube	NVIDIA RTX GPU	45 min
`Lab-I4-isaac-waypoints`	Waypoints	NVIDIA RTX GPU	30 min
`Lab-I5-isaac-line-follower`	Line follower	NVIDIA RTX GPU	30 min
`Lab-I6-isaac-ramp-imu`	Ramp and IMU	NVIDIA RTX GPU	30 min

"C++ compiler" = Visual Studio Build Tools with the *Desktop development with C++* workload (includes a Windows SDK). Labs without it are pure Python. The Isaac Sim labs need about 15 GB of disk for Isaac Sim, installed once into `C:\ix` and shared by all six.

Good to know

- **Where things go.** Each lab's ROS environment is kept in `C:\rlabs` and its build files in `C:\rb`, not inside the cloned folder (this avoids Windows' 260-character path limit and lets you move or zip a lab folder freely). pixi shares downloaded packages between labs, so the second lab is much quicker than the first.
- **One lab at a time.** Labs use the same ROS network; stop everything (Ctrl+C) before starting another lab. To clear stuck programs: `Get-Process | Where-Object { $_.Path -like "C:\rlabs\*" -or $_.Path -like "C:\ix\*" } | Stop-Process -Force`.
- **Red text in PowerShell** that starts with `Pixi task (...)` is pixi's status line, not an error.
- **Self-test.** Each lab can replay its own manual: `powershell -NoProfile -File tools\replay.ps1 README.md` (from inside the lab folder; it opens windows, so leave the mouse alone).

- **PDFs.** `pdf\` has every manual as a PDF and `ROS2-Labs-All.pdf` with all of them (rebuild with `python tools\make_pdfs.py` after `pip install markdown`).
- **What was tested, and what was not:** see TESTED.md.